



MMA Companion

Computer Science NEA

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Feedback received

Here is a list of the feedback I was given on my prototype presentation and the resultant design changes I am going to do as a result.

Feedback received	Resultant design changes
Can you filter so you can just see fighters from a club?	Filter based off club
Can the user see upcoming fights?	No design change because this is outside of the scope of my project (my project is made for looking at past stats)
Can the user update their preferences?	Allow preference update
Can a user identify a favourite fighter?	Let user set favourite fighter and let the favourite fighter affect preference scores
Do the stars at signup change the leaderboard filtering?	Preference scores/preference-based filtering
Will you show the dates of fights for fighters?	Show fight dates off events table (should've been done anyway)
Can the fights be sorted for most recent?	Filter based on fight date (linked to latter)
Can the drop down be changed so you can't get the same fighter in the odds calculation?	Change odds query to redact same fighter from list in drop down boxes on odds calculators
Can you add more stats to the fighter page (height/reach)	Add more height/reach to fighter profiles

Objective amendment

I have chosen to remove the UFC scraper since the package I thought I could use to implement the feature is now deprecated. I think that if I were to keep it in it would make the project quite a bit more unstable which isn't worth it due to the fact that it's just meant to be a demo feature.

List of design changes

This is a list of all the features I am going to add/change whether they come from development (internal), or feedback (external).

Design change	Type	Justification
Filter based off club	External	This is so the user can see who a fighter trains with which can be very important contextually
Allow preference update	External	This allows people to change their opinions if they change over time which is very likely.
Let user set favourite fighter and let it affect preference scores	External	This improves entertainment options because a user is going to be a lot more likely to find a fight with their favourite fighter in it way more entertaining
Preference scores on leaderboard/preference-based filtering	External	This lets the user look at fighters which are more likely to be important to them and with the other design changes see when they are next going to fight.
Show fight dates off events table (should've been done anyway)	External	This is so the user can see when an event happened to increase the context of the data
Filter based on fight date (linked to latter)	External	This is so the user can see the most recent/least recent fight
Change odds query to redact same fighter from list in drop down boxes on odds calculators	External	This is because it's unnecessary to see what will happen if a fighter fights themselves.
Add more height/reach to fighter profiles	External	This is to add more data for the user to look at to improve the end user experience. This also includes adding this to data population features.
Image display subroutine with threaded http request	Internal	This is not only because I need to focus on how I display fighter images, but I also need to thread the HTTP requests to stop the whole program stopping when an image takes a while to load
New odds calculation formula	Internal	The new odds calculation formula I want now considers volatility, rating deviation and inactivity
Tooltip for user IDs on modify fighter pages	Internal	Helps with data integrity but doesn't compromise usability as bad
Pie chart simplification	Internal	The pie chart algorithm now takes off 0% values to reduce clutter.

Restructured processing systems

Here I have restructured the input-processing-output tables that I already outlined in my design. I haven't put all of them here because some of them outline what they add very clearly anyway.

Sub-Problem	Input	Process	Output	REF
<u>PPROD 1: Filter on leaderboard based off of club</u>				
Open combo box off leaderboard that has a list of all the distinct clubs	User clicks on header of gym column	SELECT DISTINCT query on gyms (and include any gym option on list when populating combo box), model to table	A list of all distinct gyms for the user to search through	PPROD 1.1
When the user selects an option off the combo box, filter by fighters who go to said gyms	User clicks on a gym	SELECT query of fighters WHERE the gym is the gym the user selected, model to table	A list of fighters who only go to a certain gym	PPROD 1.2
When the user selects the "any gym" option, reset the leaderboard	User clicks on any gym option	Normal selection query that is used, model to table	Leaderboard back to normal	PPROD 1.3
<u>Allow preference update (links back to 10.5 but without the signup parts, they're done beforehand) PPROD 3:</u>				
The user can click a redo survey button and redo the survey	User clicks redo survey button	Navigate to survey page	10.5	PPROD 3
If the user has done the survey before, do an update query instead	User clicks submit button	Query to check if a survey response with the user id exists, if so then the query will be an update instead of an insert	Query sent, navigate to leaderboard	Addition to 10.5
<u>Let user set favourite fighter and let it affect preference scores:PPROD 4 (Changes 10 and 15.2)</u>				
When the user opens the survey, populate the favourite fighter combo box with a list of all fighters	User navigates to survey	SELECT names and ids from fighters table, For each result populate combo box with ids hidden	Favourite fighter dropdown box with list of possible favourite fighters	Addition to 10

When the user clicks submit on the survey, save favourite fighter to user info	User clicks submit	Along with the preference scores on the insert query, add the fighterID of the favourite fighter.	Favourite fighter declared on system.	Addition to 10
Adding favourite fighter to entertainment score calculation, also giving it a new formula to add width to take it into account	Match up rates in a list Methods in fights table Check if user's favourite fighter is one of the ones being rated	For each quality, if the number of method wins is more than average, apply this formula $newRating = oldRating - 6.25(5 - x)$ X being the number of stars the user answered to the question about the method of winning on the survey. Then if one of the fighters they're calculating about is their favourite, set x to 5 and do it one more time. If they're not, make x=2.	Display the entertainment score underneath the odds	Change to 15.2
<u>Preference scores on leaderboard/preference-based filtering: PPROD 5 (Amends 2)</u>				
When you receive the query for the leaderboard, add on survey ratings.	Result of leaderboard query	Go row to row with each fighter id and apply the calculations in done in 15 (calculate averages, make exemplar qualities apply to rating) then model the table.	Display approval score on leaderboard table	Addition to 2
<u>Show fight dates off events table (should've been done anyway): PPROD 6 (Changes</u>				
Include fight date in fights query				Change 6.2
<u>Filter based on fight date (linked to latter): PPROD 7 (Addition to 6)</u>				
When the user clicks on the fight date header, do another query that orders by date	User clicks on fight date header	Do another query of the fights but this time order by the fight dates	Ordered fights table	Addition to 6
<u>Add more height/reach to fighter profiles: PPROD 8</u>				
When the user opens the profile, display height and reach	User opens fighter profile	Display height and reach off query	Height and reach displayed	PPROD 8.1
Let the admin edit the reach and height on their side	Admin clicks increment/Decrement height/reach	If the user clicks the increment button increment the inches, let it roll onto feet if 11 inch / 0-inch threshold crossed. Don't	New height and reach change ready to submit	PPROD 8.2

	button (+ and -)	let the number go down if the values 4'10" or 7'5" are hit.		
When the admin clicks submit, update the height value and reach value	Admin clicks submit on the fighter profile	When the admin clicks submit run an SQL update query which includes the updating of the reach and height	New height and reach on system	PPROD 8.3
<u>Image display subroutine with threaded http request: PPROD 9</u>				
If the fighter's profile URL is blank, display off a photo of a default picture, if not run a http request	User/admin does an input which brings up a fighter's image	If the URL doesn't exist, set image widget to a bitmap of the file of a default profile picture. If not, run a threaded http request, turn the response into a bitmap (if there is no response get the default) and display onto the image widget	Image displayed on widget	PPROD 9.1
<u>New odds calculation formula: PPROD 10</u>				
Get both fighter's Elo info and apply new formula	Both fighter's elo and rating deviations	$E_A = \frac{1}{1 + 10^{\frac{-8(\sqrt{RD_A^2 + RD_B^2} N(A-B))}{400}}}$ Apply this formula to both fighters and output the result	New, better odds	PPROD 10.1
<u>Tooltip for user IDs on modify fighter pages: PPROD 11</u>				
Find fighter's name when admin is hovering on a fighter's ID	Admin hovers on fighter ID	SQL SELECT query to figure out fighter name based off ID Display on tooltip	Tooltip with fighter name whenever an admin hovers on an ID	PPROD 11.1
<u>Pie Chart Value elimination (addition to pie chart subproblem): PPROD 12</u>				
Simplify the pie chart	Total method wins/losses and their labels	Append a method and its label if it equals 0 before displaying onto pie chart	Cleaner looking pie chart that doesn't show valueless labels	PPROD 12.1

Restructured data

Data	Amendments
Fighter table	Adding height and reach to fighter data.
(Not in database, but generated and displayed in table model) Fighters table	Adding percentage preference score to fighter table.
Users table	Adding favourite fighter column, which is a foreign key.

Here is the fighter table's new columns.

Field name	Datatype	Description	Example	Validation	Key
Height	Varchar	Height of the fighter in feet	6'2"	Format and range	
Reach	Varchar	Reach of the fighter in feet	6'2"	Format and range	

*Both height and reach will be validated with a format check to see if they adhere to "foot'inches" and a range check from 4'10" to 7'5"

I would like to also mention that I have set Approvals.Rating to a bit instead of a Boolean but that's just a clarification, nothing else changes.

I have also added this column to the user's table

Field name	Datatype	Description	Example	Validation	Key
favouriteFighter	Int	FighterID of favourite fighter	74	Lookup	Foreign